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A social commerce purchasing decision model with trust network and item review information

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ABSTRACT

Shopping without much experience on target items is not unusual in social commerce (s-commerce). Inexperienced users are often influenced by user reviews when making purchasing decisions, which can be easily distorted due to low review quality. To address this issue, this study integrates two important review dimensions into purchasing decision model, i.e. the trust relationships to experienced users and the reliability of experienced users' reviews. A purchasing decision model for s-commerce is proposed based on trust computation and item reviews. The trust degree of inexperienced users to experienced ones is computed based on item reviews and propagated using the parameterized Hamacher t-norm operator. Based on the item review information, inexperienced users can make the final purchasing decisions on target items with trust information to experienced users and their reliability on item reviews. A numerical example with a data subset of Epinions.com is provided and the comparative analysis is conducted to verify the effectiveness of the proposed method. This study proposes an inexperienced purchasing decision method considering the risk attitude of users with flexible trust propagation parameters, which can provide a new perspective for the s-commerce recommendations.

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1. Introduction

Recently, a new commerce paradigm called social commerce (s-commerce) has emerged from the rapid growth of social media and Web 2.0 [1]. S-commerce generally refers to the utilization of social interactions among consumers for enhancing customer satisfaction and loyalty [2]. In the s-commerce environment, the demand for recommendations continues to increase [3]. However, there are few recommendation models designed for s-commerce, and most of them only consider general social relationships or item reviews, leading to low accuracy of the recommendation system [4]. From the perspective of users, this paper proposes a purchasing decision model that comprehensively considers trust relationships and item review information, which has some enlightenment for improving the precision of the group recommendation model.

One main point in e-commerce is that users with limited knowledge of target items are often frustrated with making purchase decisions [5]. We label such users as inexperienced users (IUs). Correspondingly, users who have direct experience with the target items are labeled as experienced users (EUs). It is known that online reviews are very useful for IUs to purchase unfamiliar items. Lu et al. [6] surveyed the development of e-commerce recommender systems, including recommendation models based on item reviews. Zhang et al. [7] presented a fuzzy approach for e-commerce purchasing decisions with item reviews. Najafabadi and Mahrin [8] found that the quality of comments significantly affects the accuracy of recommendations. Unfortunately, reviews are not always reliable since users sometimes can be maliciously manipulated [9,10]. Therefore, it is important to identify trustworthy EUs and provide reliable comments for IUs to make better purchasing decisions in s-commerce.

In almost all social relationships, trust is one of the most important factors that influence users' behavior in network activities since it can reduce information overload, uncertainties, and risks from unreliable reviewers [11]. The usefulness of trust can be reflected in its research in the areas of decision-making and recommendation.

(1) In the area of decision-making, trust relationships have been widely studied to deal with tough problems. To evaluate the missing information in large group decision-making, Xu et al. [12] propose a consensus model that considers the experts' trust

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relationships based on social network analysis and preference risks. Capuano et al. [13] proposed a new influence-guided fuzzy group decision-making model based on interpersonal trust with incomplete information. Wu et al. [14] developed a visual interaction consensus model for social network group decision-making with trust propagation. Wu et al. [15] proposed a social trust network partition model for large group decision-making based on trust information. In short, as Dong et al. [16] argued, the consensus reaching research with trust relationships has become one of the main topics of social network group decision-making.

(2) In the field of recommendation, trust has become an important factor to improve user satisfaction [17]. In fact, with the emergence of the social network, it was found that the general social relationships have a positive impact on the recommendation effect [18,19]. However, with the deepening of research, it has been found that trust, a more specific social relationship with stronger emotions, has a more obvious impact on the behavior of users, and existing studies have proved that trust can improve recommendation accuracy [20,21]. Yu et al. [22] proposed an adaptive trust-aware recommendation model based on a new trust measurement method. Leong et al. [23] examined the determinants of trust in s-commerce based on social presence and social support. Wan et al. [24] developed a novel trust-aware recommendation approach based on deep learning to alleviate the initialization dependence. Besides, the appearance and development of s-commerce have in turn promoted the research of trust theory and its application in recommendation [25].

The research of trust theory is essential for s-commerce due to it is unrealistic for IUs to directly contact all target EUs due to the limited scope of people's social circle. According to trust propagation [26], IUs and EUs can be connected by their mutual friends, and we call these mutual friends middle users (MUs). Since trust propagation is key for indirect trust-building, Zhang and Liu [27] proposed a social recommendation model combining trust propagation and sequential behaviors. Jiang et al. [28] extracted a domain-aware trust network to achieve more accurate trust propagation in large-scale heterogeneous trust networks. Zhao et al. [29] developed a trust mechanism based on trust transfer theory in the context of customer-to-customer s-commerce.

The trust relationship is used to help IUs find EUs, and the item review information is the key factor to help IUs make the purchasing decision. Guo et al. [30] introduced a novel recommendation model regularized with user trust and item reviews but did not consider the credibility of the evaluation information. As mentioned above, unreliable reviews can be caused by both subjective malice and non-subjective behavior. In the latter case, item ratings or recommendations from trusted friends may also be unreliable due to limited user knowledge and differences in user preferences [31]. Pranata and Susilo [32] investigated the credibility of the most popular users in giving trustworthy ratings on Yelp, a popular consumer reviews platform, and found that the most popular users are not always trustworthy in providing their ratings and their percentage of rating trustworthiness. Oh et al. [33] proposed an algorithm that iteratively adjusts a reputation based on the confidence of customer ratings to provide trustworthy reputations. Thus, we need to consider both the trustworthiness of EUs and the credibility of item reviews given by EUs.

To help IUs to buy target items in s-commerce, we propose a purchasing decision method with the trust network and reliable item review information. Firstly, we process item reviews to analyze trust relationships quantitatively and compute trust levels using Hamacher t-norm. Secondly, we evaluate the item review of IUs to the target item with reliable history reviews. Finally, IUs can make a purchasing decision with the evaluated trust value of EUs and item review of the target item. An illustrative example from a data subset of Epinions.com and the corresponding comparative analysis is provided to show the availability of the proposed method.

This paper is organized as follows: Section 2 briefly introduces the fundamentals of the proposed model. Section 3 introduces the important elements of the proposed model, including some assumptions of trust relationships, trust propagation operators, and the aggregation of reviews. Section 4 describes the steps of the proposed method in detail. Section 5 illustrates the application of the proposed method on Epinions.com. Section 6 discusses the conclusions.

2. Preliminaries

2.1. The definition of trust network

A trust network is a special kind of social network, which is mainly focus on the trust relationships between social entities. According to the basic definition of general trust networks [34], the trust network of users can be defined as:

Definition 1. A directed and weighted user trust network is defined as a triple $G = (V, E, T)$, where $V = \{v_1, \ldots, v_m\}$ is the set of users, $E = \{e_{ij}\}$ is the set of trust relationships among users, $T = \{t_{ij}\}$ is the set of trust values corresponding to E, and the general element in T represents user v_i trust v_j with trust value t_{ij}.

2.2. Hamacher t-norm

Hamacher t-norm is a generalization of many other t-norms with a parameter γ, such as algebraic and Einstein t-norms. Since the parameter γ can express the preference or risk attitude [35], or utility of DMs [36], Hamacher t-norm can be flexibly used in real applications.

A parameterized Hamacher t-norm [35] is denoted as

$$T(x, y) = \frac{xy}{\gamma + (1-\gamma)(x + y - xy)} \gamma > 0 \quad (1)$$

It is easy to verify that the parameterized Hamacher t-norm satisfies the following properties of classic t-norm [37]:

(1) Idempotent: $T(1, x) = x$, for all x;
(2) Commutative: $T(x, y) = T(y, x)$, for all x and y;
(3) Associative: $T(x, T(y, z)) = T(T(x, y), z)$, for all x, y, and z;
(4) Monotonicity: If $x \leq x_0$ and $y \leq y_0$, then $T(x, y) \leq T(x_0, y_0)$.

The parameterized Hamacher t-norm can reduce to algebraic t-norm when $\gamma = 1$, reduce to Einstein t-norm when $\gamma = 2$. Tang et al. [37] associated the meaning of the parameter γ with the risk attitude of DMs, i.e., a DM is risk-seeking when he or she prefers small γ, while the DM is risk-averse if he or she prefers large γ.

2.3. The weighted OWA operator

The weighted ordered weighted averaging (WOWA) operator [38] is an extension of the ordered weighted averaging operator (OWA), and it was proposed as a generalization of both the OWA and the weighted mean. From a practical point of view, this operator considers both the weight of information sources and the relative importance of values [39].

Definition 2. Assuming P and w are two weighting vectors of dimension n, then a mapping $f_{WOWA}: R^n \to R$ is a WOWA operator of dimension n if

$$f_{WOWA_{P,w}}(a_1, a_2, \ldots, a_n) = \sum_{j=1}^{n} w_j a_{\sigma(j)} \tag{2}$$

where σ is defined as in the case of OWA (i.e., $a_{\sigma(j)}$ is the jth largest element in the collection $a_1, a_2, \ldots, a_n$); and the weight ω_j is defined as:

$$w_j = \lambda^*\left(\sum_{i\le j} P_{\sigma(i)}\right) - \lambda^*\left(\sum_{i<j} P_{\sigma(i)}\right) \tag{3}$$

where $P = (p_1, p_2, \ldots, p_n)$ is the related n dimensional weighted vector with a_j, and $p_j \in [0, 1]$, $\sum_{j=1}^{n} p_j = 1$; λ^* is a monotone increasing function that interpolates the points $(0, 0)$ and $\left(\frac{j}{n}, \sum_{i\le j}^{n} \lambda_i\right)$.

Theorem 1. *For the n dimension f_{WOWA}, if $P = \left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}\right)$, then the WOWA operator degenerates into the OWA operator.*

The computation of weight $\boldsymbol{\omega}$ in WOWA is a key problem. To compute the weight $\boldsymbol{\omega}$, Sang et al. [39] designed the general parametric weighting function expressed as:

$$\lambda(p) = \begin{cases} 2^{r-1}p^r, & p < 0.5 \\ 1 - 2^{r-1}(1-p)^r, & p \ge 0.5 \end{cases} \tag{4}$$

For the above formula, when parameter $r = 2$ and $r = 1/2$, the parametric weighting function $\lambda(p)$ becomes the two special cases of weighting functions proposed by Torra [38]. As the reliability of reviews that EUs made on target items, P in the decision-making problem is also known. Here, we select $r = 1/2$ to compute weights $\boldsymbol{\omega}$ by Eq. (4).

3. The s-commerce purchasing decision model based on user trust and item reviews

IUs are those customers who would like to purchase a target item but have no experience either themselves or their immediate neighbors. To help IUs decide whether to buy the target item in s-commerce, we propose a purchasing decision model based on trust relationships and reliable item reviews. The decision support system for inexperienced purchasing is shown in Fig. 1. Suppose the trust information is known and we focus on how trust is computed based on item reviews and how to consider the effect of reviews' reliability on the purchasing decision.

3.1. The basic structures of the purchasing decision model

Fig. 2 shows the general framework of the purchasing decision model in s-commerce. In Fig. 2, the user–user trust network consists of trust propagation and aggregation from the IU to EUs, and the user–item review network consists of item reviews and their corresponding reliability. By combining the trust and review network, we can estimate the rating of the IU on the target item to help IU make a purchasing decision.

The user–user trust network is denoted as $G = (V, E, T)$, where $V = \{v_0, v_{ci}, v_{gj}\}(1 \le i \le m; 1 \le j \le n)$ denotes the set of users, in which, v_0 is the IU who has no purchasing experience on the target item I_0; $v_{gj}\,(1 \le j \le n)$ are EUs who have direct purchasing experience on I_0; $v_{ci}\,(1 \le i \le m)$ are MUs who have no direct purchasing experience on I_0, but can connect the IU v_0 and EUs $v_{gj}\,(1 \le j \le n)$ through effective propagation paths. $E = \{e_{ij}\}$ denotes the set of trust relationships among users, $T = \{t_{(0,ci)}, t_{(ci,gi)}, t_{(0,gi)}\}$ denotes the set of trust values. To reduce computation complexity of trust propagation, the shortest propagation paths are identified. In the propagation process, trust values $t_{(0,gi)}$ of IU v_0 in EUs v_{gj} are computed considering users' risk attitude by the Hamacher propagation operator. Finally, the average trust value $t_{(0,g)}$ of IU v_0 to all EUs is obtained.

Table 1
The representation of trust degrees.

Trust linguistic labels	The corresponding values or intervals
Complete trust	1
High trust	[0.6, 1)
Medium trust	(0.4, 0.6)
Low trust	(0, 0.4]
Complete distrust	0

The user–item review network is a heterogeneous network $P = (V, I, R, P)$, where $V = \{v_{gj}\}\,(1 \le j \le n)$ is the set of EUs, $I = \{I_0\}$ is the target item, $R = \{R_{v_{gj}\to I_0}\}$ is the set of reviews obtained from EUs on the target item I_0 and $P = \{P_{v_{gj}\to I_0}\}$ is the corresponding reliability of EUs' reviews. The reliability P is closely related with the truthfulness or rationality of the reviews on the target item I_0, which can improve the precision of a final decision. The final evaluation value $E_{v_0\to I_0}$ of the IU v_0 on I_0 is computed by the WOWA operator with multiple EUs' reviews $R_{v_{gj}\to I_0}$, in which, trust values of v_0 in EUs and the reliability P is used to compute the weights of WOWA. Finally, the IU v_0 decides whether to buy the target item I_0 according to $E_{v_0\to I_0}$ and $t_{(0,g)}$.

3.2. The assumptions and representation of trust

To determine the effective trust propagation paths between two entities, we introduce the several assumptions of trust propagation as below:

A_1: According to the "small-world effect" [40], *we assume that an initial entity can always reach to the target entity, if not, then the decision-making problem has no solution.*

This assumption ensures that the trust among users is delivered successfully despite the trust relationship is sparse.

A_2: According to the characteristics of information transferred [41], *the longer the trust propagative path, the less credibility of trust information, that is, trust suffers from attenuation metamorphic in the propagative processes.*

This assumption provides the properties of trust propagation and aggregation operator, such as the final propagative trust value is no larger than any trust value in the propagation path.

A_3 : If A trusts B, and B trusts C, then A trusts C.

This assumption is deduced from "friend of a friend should be a friend" which is generally accepted among users in social networks [25].

Under an uncertain environment, many decision-making problems are settled with linguistic variables representing decision makers' preference information [42–44]. Similarly, we represent trust level with linguistic labels and the corresponding values or intervals shown in Table 1 to make it easier to understand and express. Besides, compare the obtained trust values with the corresponding linguistic labels, users can perceive trust relationships in a more intuitive way.

3.3. The trust quantification process based on item reviews

In general, we might know that there are trust relationships between users, but the level of trust is unknown. The trust relationship is qualitative information, which can usually be measured by preference similarity [45], that is, the implicit trust level between users of social e-commerce can be determined based on

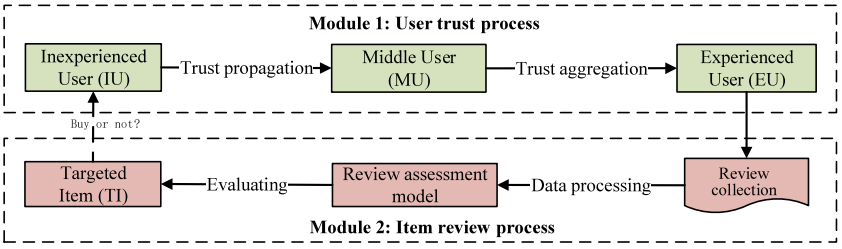


Fig. 1. The decision support system for inexperienced purchasing.

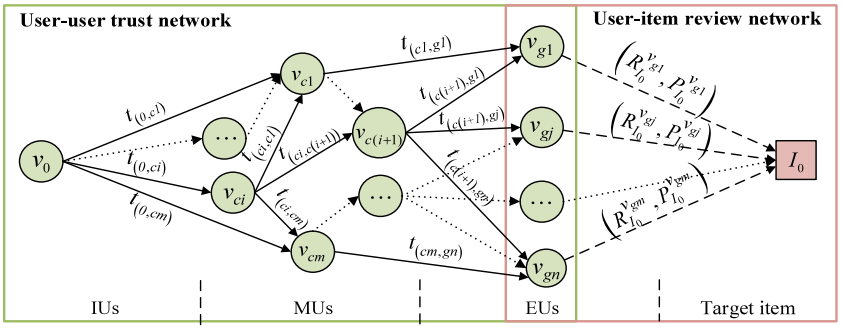


Fig. 2. The general framework of the purchasing decision model.

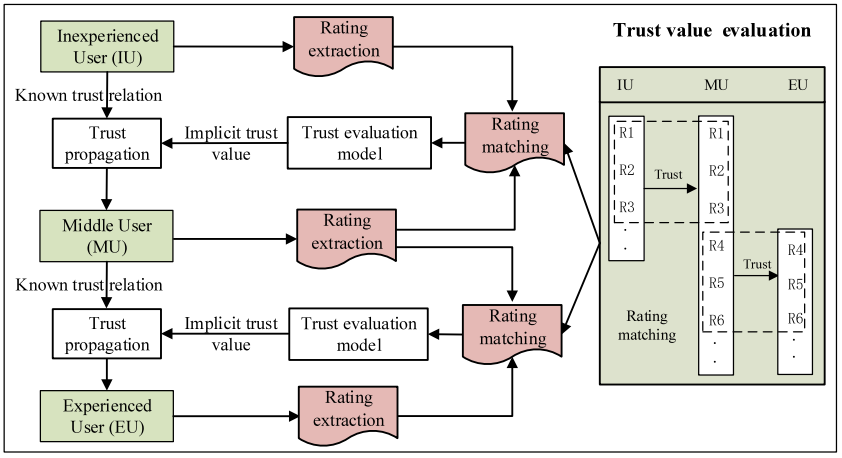


Fig. 3. The process of trust value evaluation.

their review similarity. The trust quantification process based on item review information is shown in Fig. 3.

When measuring trust levels based on item review information, the first step is to determine the number of common items rated by users. Let $N(\varphi)$ denotes the number of common items reviewed by pairwise users, if $N(\varphi)$ is too low, the trust value between the pairwise users is falsely high, otherwise, the calculation of trust level between users faces the challenge of data sparsity. To improve the precision of trust computing, we just consider the situation when the number of common items is higher than average. The average number of common items $ave(N(\varphi))$ reviewed by the pairwise users is determined as:

$$ave(N(\varphi)) = \frac{\sum N(\varphi(x,y))}{\sum N(x,y)} \tag{5}$$

where $\varphi(x,y)$ means the common rating items by all pairwise users, $N(x,y)$ means the number of the pairwise users who have reviewed at least one common item.

Trust values between users can be computed with the similarity of item reviews, in which the number of users' common items is larger than or equal to $ave(N(\varphi))$. The implicit trust values among users in this study are computed based on the generalized Jaccard similarity method [46]. For instance, for the trust value $t_{(x,y)}$ between v_x and v_y is computed as:

$$t_{(x,y)} = \alpha \times \frac{\sum_{R(\varphi(x,y))} \min\left(R_{x\to\varphi(x,y)}, R_{y\to\varphi(x,y)}\right)}{\sum_{R(\varphi(x,y))} \max\left(R_{x\to\varphi(x,y)}, R_{y\to\varphi(x,y)}\right)} + (1-\alpha) \times \frac{N(\varphi(x,y))}{\max(N(\varphi))} \tag{6}$$

where $\varphi(x,y) = I(v_x) \cap I(v_y)$, $I(v_x)$ and $I(v_y)$ represent the item set that user v_x and v_y reviewed, respectively, so $\varphi(x,y)$ represents the item set that user v_x and v_y both reviewed. $R(\varphi(x,y))$ means the ratings of items in set $\varphi(x,y)$, and $R_{x\to\varphi(x,y)}$ means the rating of v_x on these items. $N(\varphi(x,y))$ means the number of items in $\varphi(x,y)$, φ is the set of items all pairwise users have reviewed, and $\max(N(\varphi))$ means the maximum number of common items reviewed by all pairwise users.

The former part of Eq. (6) represents the generalized similarity between v_x and v_y. Since $\varphi(x,y)$ represents the item set that

users v_x and v_y both reviewed, the similarity between v_x and v_y is overrated. To improve the precision of trust values, the ratio between the number of items co-rated by two users and the maximum number of the co-rated items is considered in the later part of Eq. (6). A parameter α is used to balance the importance of the two parts. The larger the α, the greater the trust value, and the easier the IU to make decisions.

Under the assumption that an IU has no experienced neighbors he/she directly trust, the trust quantification process in this part mainly includes the trust calculation of the IU to m MUs and the trust calculation of m MUs to n EUs. Thus, the time complexity of the trust quantification process is $O(m \times n)$.

3.4. The propagation and aggregation process in trust networks

According to Fig. 2, the trust propagation and aggregation process is important for trust computing since some users may not be able to find out anyone they trust directly in real trust networks. The propagation characteristic of trust can build indirect paths for these users. A trust propagation operator is proposed based on the Hamacher t-norm to estimate unknown trust values in this paper.

Suppose user v_1 trust agent v_2, and agent v_2 trust agent v_3, then with the help of trust propagation, the trust score of agent v_1 in agent v_3 can be computed by the parameterized Hamacher product which is shown in Eq. (1):

$$t_{13} = T(t_{12}, t_{23}) = \frac{t_{12} \times t_{23}}{\gamma + (1-\gamma)(t_{12} + t_{23} - t_{12} \times t_{23})} \gamma > 0 \quad (7)$$

where t_{12} is the trust score of agent v_1 in agent v_2; t_{23} is the trust score of agent v_2 in agent v_3; $t_{12} \in [0, 1]$, $t_{23} \in [0, 1]$, t_{13} is still belongs to [0,1].

For Eq. (7), $t_{13} = t_{12} \times t_{23}$ when $\gamma = 1$. In real life, an agent may trust another agent through more than one intermediary, which means that there will be more than two arguments in Eq. (7). For example, let $v_0 \to \cdots \to v_{ci} \to \cdots v_{gj}$ is the jth trust propagation path from the IU v_0 to the EU v_{gj}, the trust value of user v_0 to v_{gj} can be computed by the generalized form of Eq. (6):

$$t_{(0,gj)} = t_{(0,c1)} \times \cdots \times t_{(ci,ch)} \times \cdots \times t_{(ck,gj)} \quad (8)$$

where $t_{(0,c1)}$ is the trust value of IU v_0 to MU v_{c1}, $t_{(ci,ch)}$ is the trust value of MU v_{ci} to v_{ch}, and $t_{(ck,gj)}$ is the trust value of MU v_{ck} to the EU v_{gj}. Similarly, we can get a generalized form of Eq. (1) when the parameter γ takes another value.

Based on Eq. (8), it is easy to prove that the trust propagation operator satisfies the following properties:

(1) $T(1, 1, \ldots, 1, t_{(ck,gj)}) = t_{(ck,gj)}$, which means that all users in the jth trust path completely trust others except v_{ck};

(2) $T(0, 0, \ldots, t_{(ck,gj)}) = 0$, which means that as long as there is a situation of "Almost no Trust" in the jth trust path, then the trust propagation is interrupted and the trust path is invalid;

(3) $T(t_{(0,ci)}, T(t_{(ci,ck)}, t_{(ck,gj)})) = T(T(t_{(0,ci)}, t_{(ci,ck)}), t_{(ck,gj)})$, which means that the trust calculation is independent of the uses' location on the trust path.

(4) Monotonicity: $t_{(0,gj)} \le \min(t_{(0,ci)}, t_{(ci,ck)}, t_{(ck,gj)})$ since $t_{(0,ci)}$, $t_{(ci,ck)}$, $t_{(ck,gj)} \in [0, 1]$, which means that the trust value is decreasing with the extension of trust path length.

Due to the large scale of users in s-commerce, the trust propagation paths among users can be very complex. In this study, we detect the shortest propagation paths between the IU and corresponding EUs [47]. Suppose there are s shortest trust propagation paths between the IU v_0 and the jth EU v_{gj} ($j = 1, 2, \ldots, n$), then the trust value $t_{(0,gj)}$ of v_0 on v_{gj} can be computed based on the average operator:

$$t_{(0,gj)} = \frac{1}{s}\left(t^1_{(0,gj)} + \cdots + t^s_{(0,gj)}\right) \quad (9)$$

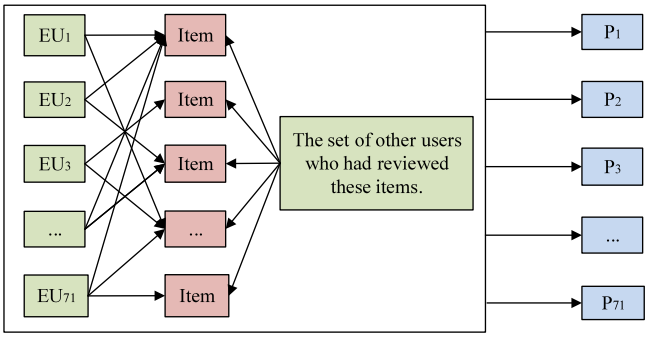


Fig. 4. The process of the reliability evaluation of item reviews.

Besides, the comprehensive trust degree $t_{(0,g)}$ of v_0 in all EUs v_g can be computed based on the average operator:

$$t_{(0,g)} = \frac{1}{n}\sum_{j=1}^{n} t_{(0,gj)} \quad (10)$$

In practice, the comprehensive trust degree $t_{(0,g)}$ implies the possibility that v_0 will make purchasing decisions according to the reviews of v_g.

3.5. The reliability determination of item reviews

In this section, we compute the reliability of item reviews given by EUs based on their ratings in all items which is shown in Fig. 4.

Firstly, we determine the item set that all EUs have reviewed, and for each of the item, find out all the corresponding ratings. Then, count the frequency of each rating value for the above items in the rating data. Finally, compute the reliability of EUs on giving item reviews.

For example, for the reliability of the EU v_{gj}, the items v_{gj} have reviewed is denoted as $I(v_{gj})$. Similarly with trust effectiveness, when the total frequency is too low, the reliability obtained is falsely high. To improve the precision of reliability computing, we just consider the situation when the total frequency is higher than average. The average frequency $aveN(I(v_{gj}))$ of all the items v_{gj} have reviewed is determined as:

$$aveN(I(v_{gj})) = \frac{\sum_{I_e \in I(v_{gj})} N(R_{v_{gj} \to I_e})}{N(I(v_{gj}))} \quad (11)$$

where $N(R_{v_{gj} \to I_e})$ denotes the rating frequency of each item I_e in $I(v_{gj})$, $N(I(v_{gj}))$ denotes the number of items in $I(v_{gj})$.

For an item $I_{e'}$ $(N(I_{e'}) \ge aveN(I(v_{gj})))$ in $I(v_{gj})$, the ratings of $I_{e'}$ given by all users are denoted by the set of $R_{v_g \to I_{e'}}$. The reliability $P_{v_{gj} \to I_{e'}}$ of the EU v_{gj} on the item $I_{e'}$ is computed as:

$$P_{v_{gj} \to I_{e'}} = \frac{N(R_{v_g \to I_{e'}})}{N(R_{all \to I_{e'}})} \quad (12)$$

where $N(R_{v_g \to I_{e'}})$ is the frequency of rating $P_{v_{gj} \to I_{e'}}$, $N(R_{all \to I_{e'}})$ denotes the total frequency of ratings given by all users.

Then, the reliability of v_{gj} on other items $P_{v_{gj} \to I_e}$ can be computed. Then, the comprehensive rating reliability $P_{v_{gj}}$ of v_{gj} can be evaluated as:

$$P_{v_{gj}} = \frac{\sum_{I_{e'} \in I(v_{gj})} P_{v_{gj} \to I_{e'}}}{N(I(v_{gj}))} \quad (13)$$

where $P_{v_{gj} \to I_{e'}}$ denotes the reliability of v_{gj} on items in $I(v_{gj})$, and $N(I(v_{gj}))$ denotes the number of the items in $I(v_{gj})$.

3.6. The review aggregation considering the reliability

If there is more than one EU who has experience in purchasing the target item, the final evaluated review of IU on the target items needs to be computed by aggregation operators. Considering the degree of importance and reviews' reliability of EUs, the review aggregation operator is proposed based on the WOWA operator shown in Eq. (3) as:

$$E_{v_0\to I_0} = f_{WOWA_{P,\omega}}\left(R_{v_{g1}\to I_0}, R_{v_{g2}\to I_0}, \ldots, R_{v_{gn}\to I_0}\right) = \sum_{j=1}^{n} \omega_j R_{v_{gj}\to I_0} \tag{14}$$

where $R_{v_{gj}\to I_0}$ is the review of EU v_{gj} on target item I_0, ω is the hybrid weight consisting of w and μ as Merigo [48]:

$$\omega_j = \beta w_j + (1-\beta)\,\mu_j \tag{15}$$

where w is the original weight of WOWA, β is set as 0.6 to highlight the importance of reviews, and μ is the weights of EUs.

To consider the importance of reviews' reliability P in the aggregation of multiple reviews, w can be obtained based on Eqs. (3) and (4) as:

$$w_j = \lambda^*\left(\sum_{i\leqslant j}^{n} P'_{\sigma(i)}\right) - \lambda^*\left(\sum_{i<j}^{n} P'_{\sigma(i)}\right) \tag{16}$$

$$\lambda\left(P'_{\sigma(i)}\right) = \begin{cases} 2^{-0.5}{P'_{\sigma(i)}}^{0.5}, & P'_{\sigma(i)} < 0.5 \\ 1 - 2^{-0.5}\left(1-P'_{\sigma(i)}\right)^{0.5}, & P'_{\sigma(i)} \geq 0.5 \end{cases} \tag{17}$$

where P' is the normalized reliability, $P'_{v_{gj}} = P_{v_{gj}} / \sum_{j=1}^{n} P_{v_{gj}}$, $\sum_{j=1}^{n} P'_{v_{gj}} = 1$, $\sigma(1), \sigma(2), \ldots, \sigma(n)$ is the substitution of $1, 2, \ldots, n$, $\sum_{j=1}^{n} w_j = 1$.

For the importance of EUs, we compute weights μ based on the trusted values of IU in EUs:

$$\mu_j = t_{(0,gj)} \Big/ \sum_{j=1}^{n} t_{(0,gj)} \tag{18}$$

where $t_{(0,gj)}$ $(j \in [1,n])$ is the trust value of the IU v_0 obtained by Eq. (9), $\sum_{j=1}^{n} \mu_j = 1$.

According to the basic properties of the WOWA operator, the review aggregation operator satisfies the following properties:

(1) Monotonicity: $f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right) \geq f_{WOWA_{P,\omega}}\left(R'_{v_{gj}\to I_0}\right)$ if $R_{v_{gj}\to I_0} \geq R'_{v_{gj}\to I_0}$. Due to that $\sum_{j=1}^{n}\omega_j = 1$, $R_{v_{gj}\to I_0} \geq R'_{v_{gj}\to I_0} \Rightarrow \sum_{j=1}^{n}\omega_j R_{v_{gj}\to I_0} \geq \sum_{j=1}^{n}\omega_j R'_{v_{gj}\to I_0}$.

(2) Commutativity: $f_{WOWA_{P,\omega}}\left(R_{v_{g\tau(i)}\to I_0}\right) = f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right)$, $R_{v_{g\tau(i)}\to I_0}$ is a reordering set of $R_{v_{gj}\to I_0}$.

(3) Idempotent: $f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right) = R$ if $R_{v_{gj}\to I_0} = R$. Since $\sum_{j=1}^{n}\omega_j = 1$, $R_{v_{gj}\to I_0} = R$, $f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right) = \sum_{j=1}^{n}\omega_j R_{v_{gj}\to I_0} = R$.

(4) Bounded: $\min_j\left\{R_{v_{gj}\to I_0}\right\} \leq f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right) \leq \max_j\left\{R_{v_{gj}\to I_0}\right\}$. Let $x_j = \min_j\left\{R_{v_{gj}\to I_0}\right\} \leq R_{v_{gj}\to I_0}$, according to the "Monotonicity" and "Idempotent", $f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right) \geq f_{WOWA_{P,\omega}}\left(x_j, x_j, \ldots, x_j\right) = x_j = \min_j\left\{R_{v_{gj}\to I_0}\right\}$; Similarly, let $y_j = \max_j\left\{R_{v_{gj}\to I_0}\right\} \geq R_{v_{gj}\to I_0}$, according to the "Monotonicity" and "Idempotent", $f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right) \leq f_{WOWA_{P,\omega}}\left(y_j, y_j, \ldots, y_j\right) = y_j = \max_j\left\{R_{v_{gj}\to I_0}\right\}$, so $\min_j\left\{R_{v_{gj}\to I_0}\right\} \leq f_{WOWA_{P,\omega}}\left(R_{v_{gj}\to I_0}\right) \leq \max_j\left\{R_{v_{gj}\to I_0}\right\}$.

4. The inexperienced purchasing decision method based on trust network and item review information

The previous section describes the main structure and solution of the proposed purchasing decision method. The main procedure of the proposed method is shown in Fig. 5.

4.1. The main steps of the proposed purchasing decision model

According to Fig. 5, the main steps of the proposed method are described as follows.

(1) Trust computation

The main purpose of this part is to obtain the trust values of IU in EUs through the trust propagation and aggregation process:

Step 1: Identify a decision-making problem in s-commerce. Suppose there is an IU v_0 tend to buy an item I_0 through trust propagation $v_0 \to v_{ci} \to v_{gj}$, $i \in [1,m]$ and $j \in [1,n]$. Except v_{gj}, neither v_0 and v_{ci} has much experience on I_0.

Step 2: Gather trust information from item reviews. According to Fig. 3, we determine the trust level from IU to MUs and MUs to EUs using Eqs. (5) and (6) based on item reviews.

Step 3: Compute the propagated trust values based on the shortest paths. Under the assumption A_4 and A_5, we firstly find the shortest propagation paths from the IU v_0 to the jth EU v_{gj}, and then compute the propagated trust values $t_{(0,gj)}$ $(j \in [1,n])$ using Eqs. (8) and (9).

Step 4: Aggregate the propagated trust values. Compute the comprehensive trust value $t_{(0,g)}$ based on Eq. (10). Finally, compare $t_{(0,g)}$ with the representations of trust degree shown in Table 1 to intuitively perceive the trust degree.

(2) Purchasing decision

The main purpose of this part is to help the IU to evaluate the target item based on the reliability of EUs' reviews:

Step 5: Compute the weights of EUs based on trust values. Compute the weights μ of EUs using Eq. (18) based on the trust values $t_{(0,gj)}$ $(j \in [1,n])$.

Step 6: Calculate the reliability of EUs on rating items. Firstly, determine the average frequency $aveN\left(I\left(v_{gj}\right)\right)$ of all the items v_{gj} have reviewed using Eq. (11). Then, compute the reliability $P_{v_{gj}\to I_{e'}}$ of the EU v_{gj} on the item $I_{e'}$ using Eq. (12). Finally, calculate the reliability $P_{v_{gj}}$ of the EU v_{gj} on rating items using Eq. (13).

Step 7: Determine the hybrid weight of the WOWA operator. Compute the original weight w for the WOWA with the normalized reliability P'_j of EUs v_{gj} based on Eqs. (16) and (17). Determine the hybrid weight ω for the WOWA using Eq. (15) with w and μ.

Step 8: Evaluate the review of IU on the target item. With the hybrid weight ω and EUs' reviews $R_{v_{gj}\to I_0}$ $(j \in [1,n])$, compute the evaluated review $E_{v_0\to I_0}$ for v_0 by the WOWA based on Eq. (14). Finally, v_0 can combine $t_{(0,g)}$ and $E_{v_0\to I_0}$ to make the final purchasing decision.

4.2. The main characteristics of the purchasing decision model

The purchasing decision model considering user trust relationships and reliable item reviews in s-commerce have been described above. This method can help customers who have no relative knowledge or experience with buying the wanted items to make purchasing decisions. For s-commerce merchants, the proposed decision method can also provide a customer-centric research perspective for recommendation models. The main characteristics of the proposed method are summarized below:

(1) The proposed method comprehensively considers the trust relationships among users and their reliable item reviews. On the one hand, it can provide credible EUs for IUs, and on the

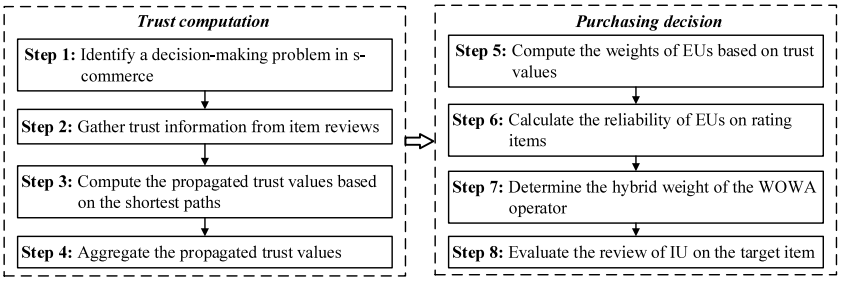


Fig. 5. The procedure of the proposed purchasing decision method.

other hand, it also considers the rating reliability of EUs from the perspective of item review.

(2) We evaluate the level of trust among users based on item reviews and consider the data sparsity problem by calculating the average of relevant user–item reviews. Besides, the obtained trust value can be compared with the trust interval defined in advance, which enables users to make intuitive decisions.

(3) In the process of trust propagation, we consider the influence of users' risk attitude on trust values based on the flexible parameter of the Hamacher t-norm operator. According to the process of the proposed decision method, the risk attitude of users is also reflected in the evaluation of item rating by IUs.

(4) EUs' rating reliability is measured based on their ratings of all items and it is considered in the computation of the weights of the WOWA operator, which can consider the importance of the trust relationship and reliable rating experience to make the decision more precise.

In short, we not only consider trust information and propagation but also focus on the reviews' reliability. The trust relationship is usually considered as the subjective judgment based on comprehensive information, while the reliability is an objective judgment based on users' rating experience. The ratings of EUs with low reliability are given low weight in WOWA aggregating, and vice versa. Also, the average trust value of IU in EUs can indicate the likelihood that the IU will make a decision based on the estimated rating. In turn, s-commerce platforms can also recommend items to IUs through this model.

5. Illustrative example

An illustrative example in Epinions.com is presented to show the application of the proposed method. Epinions.com is a kind of s-commerce platform where users can mark other users as credible persons and review items with numerical values on the scale of one to five. We illustrate the proposed model with two data sets collected by Massa [49]. One is a rating data set consisting of 49, 290 users who rated a total of 139, 738 different items with 664, 824 reviews, in which, each user rated an item at least once. The other is a trust data set which contains the trust statements issued by 49, 290 users.

5.1. Construct a data subset

Due to the large sizes of both the trust and rating data sets, it is complex to consider all users in the data set. To illustrate the proposed model with lower computational complexity, we extract a data subset with a specific IU and target item.

First, we randomly pick a user whose code is "510" from trust data as the IU iu_{510} and retrieve 149 MUs that iu_{510} trust as $MUs = \{775, 769, \ldots, 1255, 1299, 1052\}$. The longer the propagation path, the more complex the computation is, and the greater the possibility of information loss. To balance the trade-off between complexity and coverage in trust propagation, we control the reachable propagating trust distance between the IU and each EU as 2. That is, we continue to retrieve all trusted users for the MUs as the temporary EUs. According to rating data, we take the item "3571", which has not been reviewed by MUs, as an example and regard it as the target item I_{3571}. And then, match the temporary EUs and the target item in the rating data, there are 71 EUs who have reviewed I_{3571} are denoted as $EUs = \{1158, 5919, \ldots, 1836, 837, 9908\}$. It is worth noting that the trust relationships mentioned in this study are all single directions. Fig. 6 shows the structure of the data subset.

In Fig. 6, the trust relationship is given by a "1" in a binary array [0,1] which indicates a trustee trusts in the trustor only, but the specific level of trust is unknown. Trust transfer and decision inference are not available without the detailed trust values. Hence, we compute the implicit trust values through the data mining of item reviews.

5.2. The application of the proposed purchasing decision model

According to the main steps of the proposed model, the final evaluation value of user eu_{1158} on the target item I_{3571} can be computed as follows.

Step 1: Identify a decision-making problem in s-commerce

Suppose there is a decision-making problem in s-commerce with an IU iu_{510}, $MUs = \{775, 769, \ldots, 1299, 1052\}$ with 149 MUs, $EUs = \{1158, 5919, \ldots, 837, 9908\}$ with 71 EUs, and the target item I_{3571}. The overall trust decision structure of iu_{510} is shown in Fig. 6. The arrow in Fig. 6, such as $iu_{510} \rightarrow mu_{775}$ means that iu_{510} trust mu_{775}.

Step 2: Gather trust information from item reviews

The average number of common items $ave\,(N\,(\varphi))$ reviewed by the pairwise users is determined using Eq. (5) as 3.14. According to Eq. (6) with $\alpha = 0.5$, the trust value between iu_{510} and mu_{775} is computed as $t_{(iu_{510}, mu_{775})} = 0.525$. The trust values among other users can be computed similarly.

Step 3: Compute the propagated trust values based on the shortest paths

To reduce the complexity in Fig. 6, we set the length of all trust paths between v_{1071} and EUs as 2 steps and obtain the effective trust decision structure with 16 MUs and 63 EUs shown in Fig. 7, where the IU iu_{510} can reach more than 88% of EUs in the original EUs set $EUs = \{1158, 5919, \ldots, 837, 9908\}$ through 2 steps. The trust values obtained in **Step 2** are marked as the weights of arcs in Fig. 7.

Compute the trust values of v_{510}^{iu} in the EUs by the Hamacher trust propagation operator shown in Eq. (8). Also, there are more than one paths between v_{510}^{iu} and each of the EUs, the comprehensive trust values of v_{510}^{iu} in each EU is computed by the averaging operator. For example, the trust relationship between v_{510}^{iu} and v_{1158}^{eu} can be shown in Fig. 8.

According to the generalized Hamacher t-norm operator shown in Eq. (7) with the parameter $\gamma = 1$, the trust value

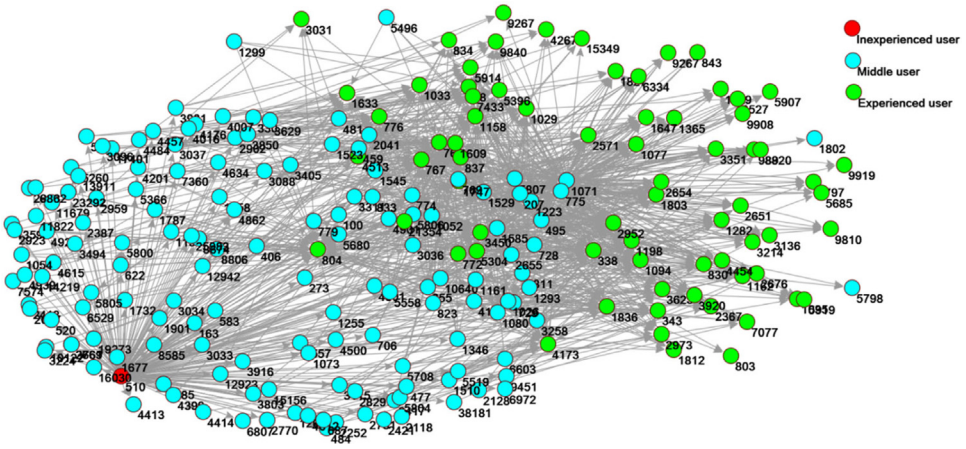


Fig. 6. The trust decision structure of iu_{510}.

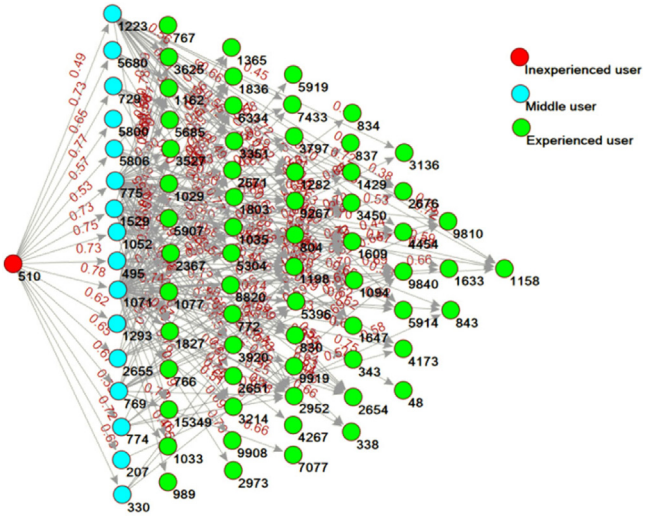


Fig. 7. The effective trust decision structure of iu_{510}.

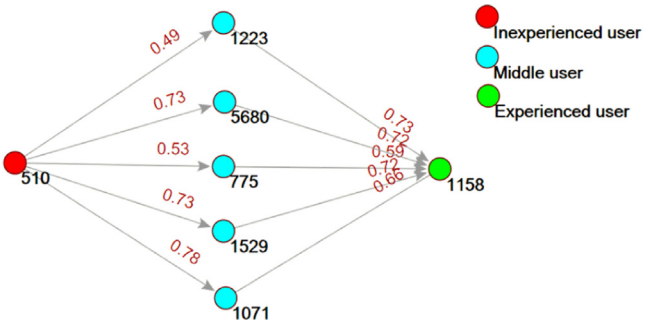


Fig. 8. The shortest trust paths between iu_{510} and eu_{1158}.

$t^1_{(iu_{510},eu_{1158})}$ of iu_{510} on eu_{1158} can be computed through the first path:

$$t^1_{(iu_{510},eu_{1158})} = 0.49 \times 0.725 = 0.355$$

The trust values of iu_{510} on eu_{1158} in other paths can be computed similarly: $t^2_{(iu_{510},eu_{1158})} = 0.523$, $t^3_{(iu_{510},eu_{1158})} = 0.311$, $t^4_{(iu_{510},eu_{1158})} = 0.526$, $t^5_{(iu_{510},eu_{1158})} = 0.517$.

Step 4: Aggregate the propagated trust values

Based on the above trust values, we can compute the comprehensive trust value $t_{(iu_{510},eu_{1158})}$ using Eq. (10):

$$t_{(iu_{510},eu_{1158})} = \frac{0.355 + 0.523 + 0.311 + 0.526 + 0.517}{5} = 0.446$$

Similarly, the comprehensive trust values of iu_{510} in other EUs can be computed. The average trust value of iu_{510} in all EUs can also be computed based on Eq. (10) with $\gamma = 1$: $t^{\gamma=1}_{(iu_{510},EUs)} = 0.440$.

Step 5: Compute the weights of EUs based on trust values

According to Eq. (18), compute the weights $\mu_{eu_{1158}}$ for eu_{1158} based on trust values:

$$\mu_{eu_{1158}} = 0.446 \Big/ \sum t_{(iu_{510},eu_j)} = 0.016$$

Similarly, the weights of other EUs can also be calculated.

Step 6: Calculate the reliability of EUs on rating items

For the reliability of the EU eu_{1158}, the items eu_{1158} have reviewed are $I\,(eu_{1158})$= {1081, 1083, 1095, . . . , 9411, 9413, 9511, 978}. The average frequency $ave\,(I\,(eu_{1158}))$ of all the items eu_{1158} have reviewed is determined using Eq. (11) as 80.22. According to Eq. (12), the reliability of eu_{1158} on all items in $I\,(eu_{1158})$ can be computed as: $P_{eu_{1158}\to I_e}$ =[0.251, 0.725, 0.304, . . . , 0.630, 0.168, 0.296]. Then, the comprehensive rating reliability of eu_{1158} can be evaluated by Eq. (13) as $P_{eu_{1158}} = 0.112$. Finally, the normalized reliability of eu_{1158} is computed:

$$P'_{eu_{1158}\to I_{1081}} = \frac{0.112}{\sum P_{EUs\to I_{1081}}} = 0.011$$

Similarly, the normalized reliability of other EUs can be computed.

Step 7: Determine the hybrid weight of the WOWA operator

Firstly, compute weights w by Eqs. (16) and (17). Then, compute the hybrid weights ω with w and μ by Eq. (15). Finally, the trust values, the related weights, the normalized reliability, and the ratings of some parts of EUs are provided in Table 2.

Step 8: Evaluate the review of IU on the target item

With the hybrid weight ω and EUs' ratings, we can obtain the estimated rating of the IU iu_{510} on the target item using the WOWA operator shown in Eq. (14) as:

$$\begin{aligned} E^{\gamma=1}_{iu_{510}\to I_{3571}} &= f_{WOWA_{P,\omega}}\left(R_{eu_{1158}\to I_{3571}}, R_{eu_{5919}\to I_{3571}}, \ldots,\right. \\ &\quad \left.\times\ R_{eu_{1836}\to I_{3571}}, R_{eu_{837}\to I_{3571}}, R_{eu_{9908}\to I_{3571}}\right) \\ &= 5 \times 0.020 + 5 \times 0.364 + \cdots + 2 \times 0.017 \\ &\quad + 2 \times 0.017 + 1 \times 0.008 \\ &= 3.942 \end{aligned}$$

Table 2
The trust values and related weights of EUs with the parameter $\gamma = 1$.

EUs	Trust values	Weights μ	Normalized reliability	Weights w	Weights ω	Ratings
1158	0.466	0.016	0.011	0.004	0.007	5
5919	0.604	0.022	0.021	0.047	0.042	4
2676	0.422	0.015	0.017	0.029	0.026	3
…						
1836	0.478	0.017	0.017	0.006	0.008	2
837	0.417	0.015	0.020	0.007	0.009	3
9908	0.541	0.020	0.013	0.005	0.008	3

Table 3
The trust values and related weights of EUs with the parameter $\gamma = 2$.

EUs	Trust values	Weights μ	Normalized reliability	Weights w	Weights ω	Ratings
1158	0.406	0.016	0.011	0.004	0.007	5
5919	0.575	0.023	0.021	0.047	0.042	4
2676	0.380	0.015	0.017	0.029	0.026	3
…						
1836	0.439	0.017	0.017	0.006	0.009	2
837	0.378	0.015	0.020	0.007	0.009	3
9908	0.507	0.020	0.013	0.005	0.008	3

The average trust value $t^{\gamma=1}_{(iu_{510},EUs)} = 0.440$ represents the likelihood that the IU iu_{510} will make decisions based on the estimated rating $E^{\gamma=1}_{iu_{510}\to I_{3571}} = 3.942$. Compare $t^{\gamma=1}_{(iu_{510},EUs)} = 0.440$ with the trust representations shown in Table 1, we can find that $t^{\gamma=1}_{(iu_{510},EUs)}$ locates in the linguistic interval of (0.4, 0.6). The IU may purchase the item I_{3571} with a medium trust and an estimated rating 3.942.

Similarly, we can obtain the average trust value $t^{\gamma=2}_{(iu_{510},EUs)}$ and the estimated rating $E^{\gamma=2}_{iu_{510}\to I_{3571}}$ of the IU iu_{510} with the parameter $\gamma = 2$ in Eq. (7). According to the main steps of the proposed method, we determine the final trust values of iu_{510} to each EU and the related weights of EUs with $\gamma = 2$ in Table 3.

The average trust value of iu_{510} in all EUs is $t^{\gamma=2}_{(iu_{510},EUs)} = 0.398$ and the estimated rating of iu_{510} to the item I_{3571} is $E^{\gamma=2}_{iu_{510}\to I_{3571}} = 3.940$. Now, we have $t^{\gamma=1}_{(iu_{510},EUs)} > t^{\gamma=2}_{(iu_{510},EUs)}$ and $E^{\gamma=1}_{iu_{510}\to I_{3571}} > E^{\gamma=2}_{iu_{510}\to I_{3571}}$, which means that the IU iu_{510} is more likely to trust unfamiliar EUs and their ratings when $\gamma = 1$, i.e., iu_{510} is risk-seeking. In turn, the IU iu_{510} is wary of trusting unfamiliar EUs and their ratings when $\gamma = 2$, i.e., iu_{510} is risk-averse.

5.3. Comparative analysis

To show the importance of trust information in purchasing decisions, a comparative analysis based on the proposed model without considering the effect of reliability is given in the following parts. As mentioned above, the personalized recommendation process of online platforms is opposite to users' purchasing decision, but they have the same purpose to satisfy users. Thus, the commonly used collaborative filtering recommendation algorithm is also used to show the positive effect of trust in the recommendation.

(1) The decisions of IUs without considering reviews reliability

If we do not consider the effect of reviews' reliability, like Ref. [30], then the normalized reliability of all EUs should be $P = (1/63, \ldots, 1/63, 1/63)$. According to Theorem 1, we can know that the WOWA degenerates to the OWA operator when $P = (1/63, \ldots, 1/63, 1/63)$. We compute the weights of the OWA operator based on the "most of" principle [50]: $w' = \{0, 0, 0, \ldots, 0.032, \ldots, 0.032, \ldots, 0.013, \ldots, 0, 0, 0\}$. And the hybrid weight ω of OWA is computed with w' and μ by Eq. (15): $\omega' = \{0.014, 0.017, 0.013, \ldots, 0.014, 0.010, 0.014\}$. The final estimated value is $E'^{\gamma=1}_{iu_{510}\to I_{3571}} = 3.504$ when $\gamma = 1$ and $E'^{\gamma=2}_{iu_{510}\to I_{3571}} = 3.499$ when $\gamma = 2$. Thus, considering whether the reviews are reliable or not, the results obtained have obvious differences. The biased results indicate that the reviews' reliability does affect the estimated reviews. Similarly, the above results also show the risk attitude of the IU iu_{510} through different values of the parameter γ.

(2) The decisions of IUs based on the collaborative filtering recommendation algorithm

The collaborative filtering algorithm is widely used in recommendations [8]. Its main idea is to recommend items for IUs based on the feature matching between the IU and EUs. To compare with the collaborative filtering algorithm, we match iu_{510} and all the EUs $v = \{37843, 834, 23223, \ldots, 6076, 7433, 1029, 3136\}$ who have reviewed I_{3571}, in which, the matching degree is determined by the trust value obtained in Eq. (6).

Firstly, we compute the matched degree between iu_{510} and each user in v using Eq. (6). Then, we compute the weights of users in v with the matched degrees using Eq. (18). Finally, we compute the evaluated rating for iu_{510} by the weighted averaging operator: $E''_{iu_{510}\to I_{3571}} = 3.546$, which is independent with the Hamacher t-norm parameter and cannot consider the risk attitude of users. We can find that the evaluated rating is different from that obtained by the proposed model. The difference is mainly due to a lack of experience of trusted users and their reliability.

The importance of review reliability is verified in the first comparison, and the importance of trust relationships and review reliability both are verified in the second comparison. In practice, it is a common problem in online shopping that the decision-making precise may be affected by suspicious reviews. Unfortunately, it is not easy for IUs to identify suspicious reviews. Compared with other methods, the objectivity of the practical ratings can be judged based on the estimated rating obtained by the proposed model. For example, when the evaluated rating is significantly lower than the practical ratings, it indicates that there may be some paid praise in practical ratings. When the evaluated rating is significantly higher than the practical ratings, it indicates that there may be some malicious negative comments in practical ratings. Besides, the comparison suggests that there is some correlation between trust and preference. In a word, trust relationships can not only provide purchasing experience for IUs but also help them to filter out false positive or malicious negative comments.

6. Conclusions

Since trust propagation can build bridges for strangers and reliable reviews can provide suggestions for users in s-commerce, a purchasing decision method is proposed based on trust relationships and item reviews. According to the proposed method, IUs can decide whether to buy target items according to trusted experience.

In the proposed method, we represent trust intuitively with linguistic labels and corresponding interval values. We compute the trust degree among users based on item reviews and determine the final trust value for the IU to EUs using the parameterized Hamacher t-norm operator. In the item review process, the propagative trust values of the IU to EUs and the review reliability of EUs are both considered to evaluate the rating of IUs on the target item. We illustrate the application of the proposed model using a data subset from Epinions.com. Compared to the traditional collective filtering recommendations, we consider the role of trust in recommendation accuracy. Compare to the trust-based

recommendations, we further consider the trust quantification and review reliability.

The proposed model and results may shed some important management insights for existing s-commerce practice. First of all, the common malicious negative reviews and the paid positive reviews may mislead users' decisions. Based on the combination of trust information and review reliability, the proposed model can maximize the power of social interactions in e-commerce and reduce the negative effect of unreliable reviews on purchasing decisions. Furthermore, because of the sharing behavior of people on purchasing products, the propagative nature of trust can enhance user confidence in trying unacquainted items and thus increasing customer base. In a word, the proposed model may improve the recommendation accuracy and develop potential users for s-commerce platforms.

Since distrust is not the opposite of low-level trust, it is expected to have different effects on purchasing decisions for IUs. People may behave differently when they face distrust and low trust. In some special situations, distrust may benefit users more in decision-making than trust. Hence, it would be useful to explore the differences and connections between distrust and trust and discuss their influences on s-commerce in future research.

CRediT authorship contribution statement

Tong Wu: Methodology, Writing – original draft. **Ru Zhang:** Software, Visualization, Investigation. **Xinwang Liu:** Conceptualization, Supervision. **Fang Liu:** Data curation, Data process. **Yi Ding:** Writing – reviewing and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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